# Permutation Tests

Mark M. Fredrickson (mfredric@umich.edu)

Module 3: Resample

Computational Methods in Statistics and Data Science (DATASCI 406)

## The null hypothesis can be the simulator

A $p$-value needs the distribution of $T$ under $H_0$.

- **Module 1**: $H_0$ fixed $F$. Simulate from $F$, compute $T$, count.
- **Deck 1**: $F$ unknown. Resample from $\hat{F}$. That is the distribution of $T$ around $\hat{\theta}$, not under $H_0$.
- **Today**: $H_0$ says the data would look the same after a **rearrangement**. Rearrange, compute $T$, count.

No $F$ is needed, and the test is exact.

## The recipe: condition, rearrange, compute, count

Four things to name, every time:

1. $H_0$: some rearrangement leaves the data's distribution unchanged.
2. **Condition on** what $H_0$ says nothing about.
3. **Rearrange** the rest. Given step 2, every rearrangement is equally likely under $H_0$.
4. $T$: compute it on every rearrangement, or on $B$ random ones.

$p$-value: the fraction of rearrangements with $T$ at least the observed value.

Flint: if flushing does nothing, swapping a home's two bottles changes nothing.

Today: **flip signs** and **shuffle labels**.

## $H_0$ picks the rearrangement, the alternative picks $T$

- The **rearrangement** comes from $H_0$. It makes the level of the test exact.
- The **statistic** comes from the alternative. It decides which departures from $H_0$ the test can see.

The deck follows that split: two rearrangements, then the choice of statistic, then what changes when the null is about a parameter rather than $F = G$.

# Rearrangement 1: Signs

## Under symmetry, the signs are fair coins given the magnitudes

$D_1, \ldots, D_n$ independent, not necessarily identically distributed.

$$H_0 : D_i - \theta_0 \text{ and } -(D_i - \theta_0) \text{ have the same distribution, for each } i.$$

Each home may have its own spread. Only symmetry about $\theta_0$ is assumed.

Write $D_i - \theta_0 = S_i \, |D_i - \theta_0|$, $S_i = \pm 1$. Under $H_0$, given the magnitudes, the $S_i$ are independent fair coins.

## Flipping signs gives an exact test of symmetry

- $H_0$: each $D_i$ symmetric about $\theta_0$.
- **Condition on**: the magnitudes $|D_i - \theta_0|$.
- **Rearrange**: the signs. All $2^n$ patterns equally likely.
- $T$: $\frac{1}{n}\sum_i (D_i - \theta_0)$, the mean of the signed differences.

$p$-value: the fraction of patterns with $T$ at least the observed. The observed pattern is one of them, so $p \geq 2^{-n}$.

Nothing beyond symmetry was used.

## If flushing does nothing, $D_i$ is symmetric about 0

Deck 1: $D_i = \log X_i - \log Y_i$, first draw against after 2 minutes of flushing, in 271 homes.

$H_0$: flushing makes no difference, so the two bottles from a home are **exchangeable**: $(X_i, Y_i)$ and $(Y_i, X_i)$ have the same distribution.

Then $\log X_i - \log Y_i$ and $\log Y_i - \log X_i$ have the same distribution: $D_i$ is symmetric about $\theta_0 = 0$.

```
d <- log(vt$first_draw) - log(vt$flush_2min)
round(d[11:20], 2)
```

```
##  [1]  3.39  1.72  1.53  1.32  1.48 -0.15  1.75  0.42  1.52
## [10]  1.40
```

## Ten homes: all 1024 patterns, $p = 2/1024$

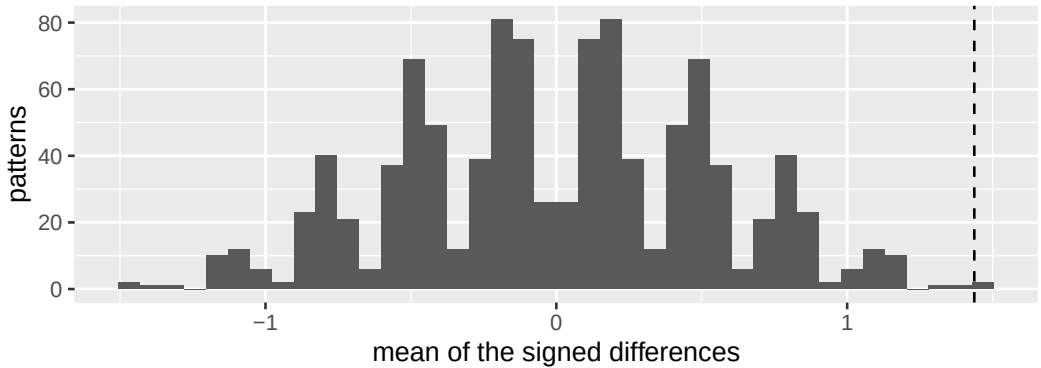
`expand.grid` lists every combination of its arguments, so each row of `S` is one of the $2^{10} = 1024$ sign patterns.

```
d10 <- d[11:20]
S <- as.matrix(expand.grid(rep(list(c(-1, 1)), 10)))
null10 <- as.vector(S %*% abs(d10)) / 10
c(observed = mean(d10), p = mean(null10 >= mean(d10) - 1e-12))
```

```
## observed        p
## 1.437092 0.001953
```

One of the ten is negative. Only the observed pattern and the one with all signs positive give $T$ that large, so $p = 2/1024$. The matrix product can round the observed pattern's mean just below `mean(d10)`, and the `1e-12` keeps it counted.

## The sign-flip null is symmetric about 0




It is symmetric about 0, since $S$ and $-S$ are equally likely.

## 271 homes: draw patterns at random

$2^{271}$ patterns are too many to list. Draw $B$ at random, and use the $(1 + \#)/(B + 1)$ $p$-value from module 1.

```
t_obs <- mean(d)
t_flip <- replicate(B, mean(sample(c(-1, 1), n, replace = TRUE) * abs(d)))
c(observed = t_obs, max_null = max(t_flip),
  p = (1 + sum(t_flip >= t_obs)) / (B + 1))
```

```
##  observed  max_null         p
## 1.4278869 0.3384585 0.0002499
```

No random sign pattern comes close, and the $p$-value is $1/(B + 1)$, the smallest possible.

## The sign test needs only a median

- $H_0$: $P(D_i > \theta_0) = 1/2$. The median is $\theta_0$.
- **Condition on**: only $n$. The signs are fair coins on their own, not given the magnitudes.
- **Rearrange**: the signs.
- $T$: the number of positive differences. It never reads the magnitudes.

So $T \sim \text{Binomial}(n, 1/2)$ whatever the magnitudes are. No enumeration, no symmetry.

Flint: 246 positive, 24 negative, one zero dropped.

```
binom.test(sum(d > 0), sum(d != 0))$p.value
```

```
## [1] 1.478e-47
```

## The less $T$ reads, the less $H_0$ must guarantee

- The sign test reads only the signs. $H_0$ need only make each sign a fair coin: the median is $\theta_0$.
- The mean reads the magnitudes too. Every sign pattern must be equally likely **given the magnitudes**: symmetry.
- The weaker null costs power. The sign test cannot see how far from $\theta_0$ the differences are.

## The wild bootstrap approximates, the sign-flip test is exact

- Wild bootstrap: flip the signs of **residuals** to approximate a sampling distribution. Approximate.
- Sign-flip test: flip the signs of $D_i - \theta_0$, fair coins under $H_0$. Exact.
- Assumes **independence** and **symmetry**. Not normality, not equal variances.
- Paired data: symmetry came from **exchangeable pairs**. Equal marginals are not enough, since the two bottles from a home are dependent.

# From Test to Interval

## Inverting the test gives an interval

Module 1's trick: keep every $\theta_0$ the two-sided 5% test does not reject. The kept values are a 95% interval.

At each $\theta_0$, $H_0$ is symmetry about $\theta_0$. The null is symmetric about 0, so compare $|T^*|$ with $|T|$.

One sign matrix serves every $\theta_0$. If row $b$ of `S` is a pattern,

$$T_b^*(\theta_0) = \frac{1}{n}\sum_i S_{bi}(d_i - \theta_0) = \underbrace{\frac{1}{n}\sum_i S_{bi} d_i}_{\text{one matrix product}} - \theta_0 \underbrace{\frac{1}{n}\sum_i S_{bi}}_{\text{row means}}.$$

## One sign matrix serves every $\theta_0$

```
S <- matrix(sample(c(-1, 1), B * n, replace = TRUE), B)
Sd <- as.vector(S %*% d) / n
Sbar <- rowMeans(S)
p_two <- function(theta0) {
  t0 <- mean(d) - theta0
  (1 + sum(abs(Sd - theta0 * Sbar) >= abs(t0))) / (B + 1)
}
grid <- seq(1.2, 1.7, by = 0.002)
keep <- grid[map_dbl(grid, p_two) > 0.05]
range(keep)
```

```
## [1] 1.290 1.564
```

## The sign-flip and bootstrap intervals nearly agree

```
boot_d <- replicate(B, mean(sample(d, replace = TRUE)))
exp(rbind(sign_flip = range(keep),
          bootstrap = quantile(boot_d, c(0.025, 0.975))))
```

```
##             2.5% 97.5%
## sign_flip 3.633 4.778
## bootstrap 3.624 4.789
```

On the original scale, flushing cuts lead by a factor of about 4.2. The two intervals nearly agree. The sign-flip interval assumes symmetry, and the bootstrap interval is approximate.

# Rearrangement 2: Labels

## Under $F = G$, every split of the pooled values is equally likely

$X_1, \ldots, X_m \overset{\text{iid}}{\sim} F$ and $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} G$, independent.

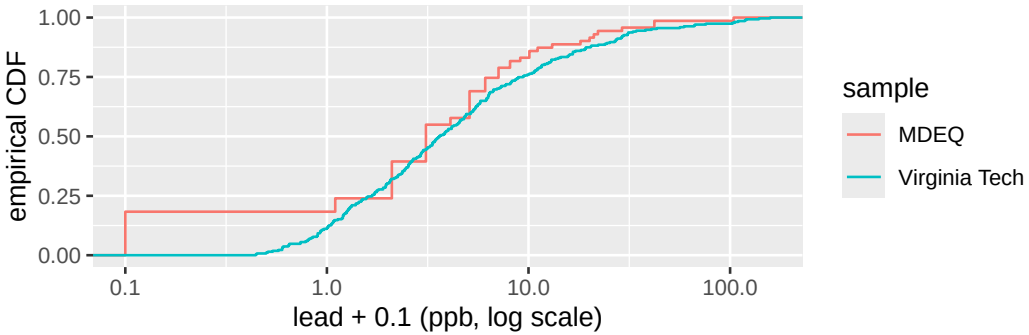
- $H_0$: $F = G$. The $N = m + n$ pooled values are then iid, and the labels carry no information.
- **Condition on**: the pooled values.
- **Rearrange**: the labels. Each of the $\binom{N}{m}$ splits into $m$ and $n$ is equally likely.
- $T$: $\bar{X} - \bar{Y}$, to start.

Shuffle the labels, recompute $T$, count.

# Did MDEQ and Virginia Tech sample the same distribution?

In 2015 MDEQ collected 71 samples, and Virginia Tech collected 271 first-draw samples.

$H_0$: the two agencies sampled the same distribution of lead, $F = G$.



## Shuffle the labels: the difference in means does not reject

g is TRUE for the MDEQ samples. sample(g) shuffles the labels and keeps the group sizes.

```
diff_means <- function(z, g) mean(z[g]) - mean(z[!g])
t_obs <- diff_means(z, g)
t_perm <- replicate(B, diff_means(z, sample(g)))
c(observed = t_obs, p = (1 + sum(abs(t_perm) >= abs(t_obs))) / (B + 1))
```

```
## observed        p
##  -3.3361   0.2302
```

# Choosing the Statistic

## The rearrangement gives the level, the statistic gives the power

$H_0$ fixes the rearrangement. Any $T$ gives an exact test from the same rearrangements.

- Signs: the mean and the count came from the same flips.
- Labels: $\bar{X} - \bar{Y}$ so far.

$T$ decides which departures from $H_0$ the test can see.

Choose it before looking. Trying several and reporting the smallest $p$-value invalidates the test.

## Means, medians and KS see different departures from $F = G$

Same $H_0$, $F = G$, and the same shuffle of labels. Only $T$ changes.

- **Difference in means**: a shift in the center. Sensitive to the largest values.
- **Difference in medians**: a shift in the middle.
- **Kolmogorov-Smirnov**, $D = \max_x |\hat{F}_m(x) - \hat{G}_n(x)|$: any difference between the two CDFs.

## Three statistics from the same shuffles

```
ks_stat <- function(z, g) max(abs(ecdf(z[g])(z) - ecdf(z[!g])(z)))
three <- function(z, g) c(mean = diff_means(z, g),
                          median = median(z[g]) - median(z[!g]),
                          ks = ks_stat(z, g))
t3 <- three(z, g)
perm3 <- replicate(B, three(z, sample(g)))
(1 + rowSums(abs(perm3) >= abs(t3))) / (B + 1)
```

```
##    mean  median      ks
## 0.21270 0.54561 0.04149
```

## KS finds the gap at 0, a difference in recording

```
v <- sort(unique(z))
gap <- ecdf(z[g])(v) - ecdf(z[!g])(v)
c(at = v[which.max(abs(gap))], gap = gap[which.max(abs(gap))])
```

```
##     at    gap
## 0.0000 0.1831
```

- The largest gap is at 0. MDEQ recorded whole numbers, 13 of them 0, and no Virginia Tech sample is below 0.344 ppb.
- The KS test rejects equal distributions, and the reason is in how the values were recorded.

## $F = G$ includes how the values were recorded

```
z_round <- c(mdeq$lead, round(vt$first_draw))
t_round <- ks_stat(z_round, g)
perm_round <- replicate(B, ks_stat(z_round, sample(g)))
(p_round <- (1 + sum(perm_round >= t_round)) / (B + 1))
```

```
## [1] 0.06923
```

Recorded the same way, the KS $p$-value goes from 0.041 to 0.069. The null is **equal distributions**, and a test that sees the whole distribution will find any difference, including one in measurement.

# Statistics You Can Tabulate

## Ranks: same $H_0$, same shuffle, a $T$ that forgets the data

Same $H_0$, $F = G$, and the same shuffle of labels. A new $T$:

1. Replace each value by its **rank** in the pooled sample, from 1 to $N$.
2. Compute $T$ from the ranks, such as the sum of the $X$ ranks.

Under $H_0$, with no ties, the $X$ ranks are a random subset of size $m$ of $\{1, \ldots, N\}$. The data no longer appear in that statement.

## A rank null depends only on $m$ and $n$, so it could be tabulated

- Every null so far depended on the data, through the magnitudes or the pooled values.
- A rank statistic's null depends only on $m$ and $n$. Compute it once and tabulate it. That is what rank tests were for before computers.
- The price: ranks ignore how far out the values are. The largest Virginia Tech sample, 158 ppb, has rank $N$, as it would at 30 ppb.

## The rank-sum permutation test matches `wilcox.test`

$W$ is the sum of the $X$ ranks. `wilcox.test` reports $W - m(m+1)/2$, the same null shifted.

```
r <- rank(z)
w_obs <- sum(r[g])
w_perm <- replicate(B, sum(r[sample(g)]))
mid <- sum(g) * (length(z) + 1) / 2
c(permutation = (1 + sum(abs(w_perm - mid) >= abs(w_obs - mid))) / (B + 1),
  wilcox = wilcox.test(mdeq$lead, vt$first_draw)$p.value)
```

```
## permutation      wilcox
##      0.2204      0.2241
```

Ties get their average rank, and `wilcox.test` then uses a normal approximation.

## Signed ranks: the same move for sign flips

Same flips and the same $H_0$, symmetry about $\theta_0$. $T$ is the sum of the ranks of $|D_i - \theta_0|$ over the positive differences, so the null depends only on $n$.

It reads the ranks of the magnitudes, so unlike the sign test it needs symmetry.

```
rk <- rank(abs(d10))
v_obs <- sum(rk[d10 > 0])
P <- as.matrix(expand.grid(rep(list(0:1), 10)))  # which are positive
v_null <- as.vector(P %*% rk)
c(exact = mean(v_null >= v_obs),
  wilcox = wilcox.test(d10, alternative = "greater")$p.value)
```

```
##    exact   wilcox
## 0.001953 0.001953
```

## KS is a rank statistic

Same $H_0$, same shuffle, $T = D$.

Sort the pooled sample. Let $a_k$ be the number of $X$s among the $k$ smallest. Then

$$D = \max_k \left| \hat{F}_m(z_{(k)}) - \hat{G}_n(z_{(k)}) \right| = \max_k \left| \frac{a_k}{m} - \frac{k - a_k}{n} \right|.$$

$a_k$ depends only on which pooled ranks are $X$ ranks: under $H_0$, with no ties, a random $m$-subset.

So the null of $D$ depends on $m$ and $n$ only, at every sample size.

`ks.test` computes it exactly for $mn < 10{,}000$. Flint has $mn = 19,241$, so it uses a limit by default.

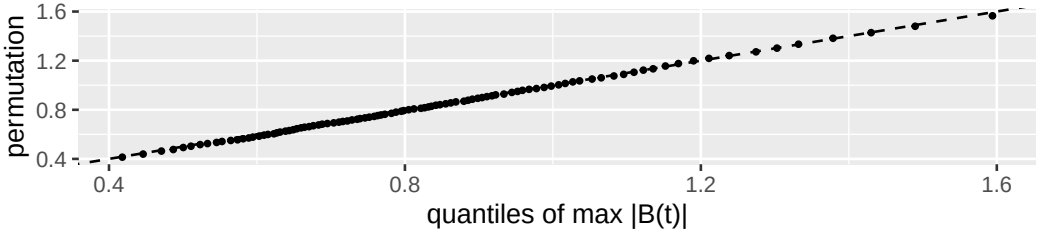
## For large $m$ and $n$, the KS null is the bridge maximum

Module 2, deck 4: $\sqrt{n}(\hat{F}_n(x) - F(x))$ behaves like $B(F(x))$. So for large $m$ and $n$,

$$\sqrt{\frac{mn}{m+n}}\, D \to \max_{0 \le t \le 1} |B(t)|.$$

```
bridge_max <- replicate(2000, {
  w <- cumsum(rnorm(1000)) / sqrt(1000)
  max(abs(w - (1:1000) / 1000 * w[1000]))
})
scaled <- sqrt(nrow(mdeq) * nrow(vt) / length(z)) * perm3["ks", ]
```

## Here the limit matches the permutation null




- The limit needs large $m$ and $n$, the rank null needs no ties. `ks.test` warns about ties unless told `exact = TRUE`.
- The permutation null needs neither. It conditions on the data, ties included, and is exact for them.
- Tabulated nulls were the way around not computing. Now any $T$ gets its null from the shuffles.

# Equality or a Parameter?

## $F = G$ is stronger than $\theta_F = \theta_G$

- The shuffle tests **equality**, $F = G$, exactly.
- The question is often a **parameter**: $\theta_F = \theta_G$, the two means, say.
- $F = G$ implies $\theta_F = \theta_G$, not the reverse. Rejecting $F = G$ does not say what differs.
- Under $\theta_F = \theta_G$ with $F \neq G$, shuffled data do not look like the original. The level is no longer exact.

## Equal means, unequal spreads: $\theta_F = \theta_G$ but $F \neq G$

Take

$$X_1, \ldots, X_{71} \overset{\text{iid}}{\sim} N(0, \sigma_X^2), \qquad Y_1, \ldots, Y_{271} \overset{\text{iid}}{\sim} N(0, \sigma_Y^2),$$

the Flint group sizes, with $\sigma_X \neq \sigma_Y$.

The means are equal and $F \neq G$. Shuffling labels mixes the two spreads.

Does the choice of $T$ rescue the test? Three candidates: the difference in means, the rank sum, and **Welch's** $t$, the difference divided by an SE from each group's own variance.

## Means, the rank sum, and Welch's $t$

```
stats <- function(z, g) {
  a <- z[g]
  b <- z[!g]
  se <- sqrt(var(a) / length(a) + var(b) / length(b))
  c(diff = mean(a) - mean(b),
    ranksum = sum(rank(z)[g]) - length(a) * (length(z) + 1) / 2,
    welch = (mean(a) - mean(b)) / se)
}
```

The rank sum is centered at its null mean $m(N+1)/2$, so all three are two-sided tests on $|T|$.

## Simulate the level of each permutation test

Each replication draws one dataset with equal means, and runs a permutation test with each statistic.

```
rejects <- function(sd_x, sd_y, m = 71, n = 271, R = 500, B = 400) {
  g <- rep(c(TRUE, FALSE), c(m, n))
  rowMeans(replicate(R, {
    z <- c(rnorm(m, 0, sd_x), rnorm(n, 0, sd_y))
    t0 <- stats(z, g)
    tp <- replicate(B, stats(z, sample(g)))
    (1 + rowSums(abs(tp) >= abs(t0))) / (B + 1) <= 0.05
  }))
}
```

## Rejection rates at nominal 5%

```
rbind(`sd_x = 3, sd_y = 1` = rejects(3, 1),
      `sd_x = 1, sd_y = 3` = rejects(1, 3),
      `sd_x = 1, sd_y = 1` = rejects(1, 1))
```

```
##                      diff ranksum welch
## sd_x = 3, sd_y = 1 0.226   0.128 0.058
## sd_x = 1, sd_y = 3 0.004   0.008 0.044
## sd_x = 1, sd_y = 1 0.050   0.056 0.048
```

With 500 replications, each rate has a Monte Carlo SE of about 0.01.

## Only the studentized statistic keeps its level

- Shuffling pools the variances: the permutation SE uses $\hat{\sigma}^2_{\text{pooled}}(1/m + 1/n)$, the truth is $\sigma_X^2/m + \sigma_Y^2/n$.
- **Small group, large spread**: the permutation SE is too small and the test over-rejects. The other way, it under-rejects.
- Ranks soften it. A bounded $T$ caps the damage, but $mn(N+1)/12$ still assumes $F = G$.
- Welch's $t$ uses each group's own variance. Its true and permutation nulls both approach $N(0,1)$, so the level holds in the limit.
- For $F = G$, any $T$. For a parameter, studentize.

# Summary

## Summary: the recipe

Condition on what $H_0$ ignores. Given it, every rearrangement of the rest is equally likely. Compute $T$ on them, count.

- **Signs**, under symmetry: $2^n$ patterns, given the magnitudes.
- **Labels**, under $F = G$: $\binom{N}{m}$ splits, given the pooled values.
- List them for small $n$, draw $B$ otherwise, and use $(1 + \#)/(B + 1)$.
- Invert the test for an interval. One matrix serves every $\theta_0$.

## Summary: the two choices

- The **rearrangement** comes from $H_0$ and sets the level: exact, for any $T$.
- The **statistic** comes from the alternative and sets the power.
- The less $T$ reads, the less $H_0$ must say: signs only need a median, magnitudes need symmetry.
- A rank null depends only on $m$ and $n$. KS is a rank statistic, and the bridge is its limit.
- A parameter null, $\theta_F = \theta_G$, is weaker than $F = G$. Ranks soften a difference in spread, a studentized $T$ fixes it in the limit.